

SAN JUAN MARIN INTL, Puerto Rico

WMO: 785260

Lat: 18.4317N

Lon: 65.9919W

Elev: 12

StdP: 14.69

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 11641

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	70.0	71.0	59.6	76.1	76.4	61.4	81.4	76.7	22.4	80.0	20.9	80.3	2.1	150	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	91.6	77.5	89.8	77.8	89.0	77.9	81.0	86.3	80.3	86.1	79.8	85.8	11.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.3	151.8	83.7	78.7	149.0	83.5	78.0	145.7	83.3	44.6	86.6	43.9	86.1	43.3	85.8	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.0	19.5	18.1	DB	67.9	93.8	2.5	1.3	66.1	94.7	64.6	95.5	63.2	96.2	61.4	97.1
			WB	62.8	82.2	2.1	1.3	61.3	83.1	60.1	83.9	59.0	84.7	57.5	85.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.8	77.5	77.6	78.4	79.8	81.3	83.0	83.1	83.3	83.4	82.5	80.4	78.7
	DBStd	2.81	1.68	1.73	1.86	1.94	2.01	1.68	1.58	1.57	1.79	1.79	1.74	1.75
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11230	853	772	881	895	971	989	1025	1032	1001	1008	912	891
	CDD65	5755	388	352	416	445	506	539	560	567	551	543	462	426
	CDH74	55282	2530	2298	3052	3894	4994	5982	6350	6503	6275	5811	4273	3320
	CDH80	16169	383	396	655	988	1414	2046	2212	2325	2264	1878	1016	592
Wind	WSAvg	7.8	7.7	8.3	8.3	8.5	8.0	8.8	9.6	8.4	6.7	5.8	6.4	7.6
Precipitation	PrecAvg	55.6	3.2	2.2	2.2	4.1	5.7	4.3	5.0	5.9	5.8	5.4	6.6	4.8
	PrecMax	89.6	11.1	6.7	9.3	15.0	15.0	13.6	14.2	18.6	14.8	15.1	17.6	16.8
	PrecMin	32.6	0.6	0.2	0.0	0.0	0.0	0.2	1.1	1.9	1.8	1.1	0.4	0.6
	PrecStd	13.2	2.0	1.3	1.6	3.1	3.7	2.9	2.5	2.8	2.9	2.9	3.5	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.7	87.6	89.3	91.3	91.7	92.3	91.6	92.0	93.1	92.1	89.0	87.6
		MCWB	73.5	73.0	72.9	75.6	76.5	76.9	77.9	78.2	78.1	77.6	77.2	75.1
	2%	DB	84.8	85.3	86.8	88.1	89.1	89.9	89.4	89.7	91.1	89.5	87.3	85.7
		MCWB	73.7	73.0	73.9	75.2	76.6	77.8	78.4	78.8	78.3	78.2	77.4	75.1
	5%	DB	83.6	84.0	85.1	86.3	87.7	88.6	88.3	88.6	89.0	88.1	86.1	84.6
		MCWB	73.7	73.2	73.7	75.2	76.9	77.9	78.3	78.7	78.5	78.1	77.0	75.2
	10%	DB	82.5	82.8	83.7	84.9	86.2	87.6	87.6	87.8	87.9	87.1	85.0	83.5
		MCWB	73.5	73.0	73.4	74.8	76.8	77.8	78.3	78.6	78.5	78.0	76.7	74.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.2	76.5	77.3	78.7	80.4	81.4	81.5	81.7	81.7	81.7	80.1	78.7
		MCDB	82.0	82.5	83.3	84.9	85.5	85.9	86.5	87.1	86.9	86.1	85.0	83.3
	2%	WB	76.0	75.4	76.0	77.4	79.3	80.1	80.4	80.7	80.5	80.3	79.1	77.3
		MCDB	81.1	81.6	82.3	83.8	84.9	85.8	86.0	86.5	87.0	85.6	84.4	82.4
	5%	WB	75.2	74.5	75.1	76.5	78.4	79.3	79.8	80.0	79.8	79.5	78.2	76.5
		MCDB	80.5	80.8	81.6	83.0	84.2	85.6	85.7	86.0	86.5	85.4	83.6	81.6
	10%	WB	74.4	73.8	74.4	75.7	77.6	78.7	79.1	79.4	79.1	78.8	77.4	75.7
		MCDB	79.8	80.0	80.7	82.3	83.4	85.3	85.4	85.7	86.0	85.2	82.8	80.9
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.8	11.2	11.0	10.5	10.5	10.0	10.1	10.5	10.6	9.9	9.7
		MCDBR	12.0	12.6	13.3	13.0	12.1	11.5	10.6	10.9	11.5	11.5	11.3	11.2
		MCWBR	4.9	4.9	4.9	4.8	4.5	4.5	4.4	4.6	4.7	4.7	4.7	4.6
	5% WB	MCDWR	10.6	11.3	11.7	11.3	10.6	10.7	10.1	10.2	10.7	10.7	10.4	9.7
MCWBR		5.1	5.2	5.1	5.0	4.8	4.9	4.8	5.0	5.1	5.0	4.8	4.9	
Clear-Sky Solar Irradiance	taub	0.372	0.387	0.387	0.414	0.455	0.529	0.528	0.491	0.444	0.393	0.387	0.378	
	taud	2.548	2.494	2.491	2.445	2.340	2.093	2.094	2.230	2.391	2.552	2.566	2.561	
	Ebn at Noon	289	290	293	284	268	247	248	259	272	285	283	284	
	Edh at Noon	31	34	35	37	41	52	52	45	38	32	30	30	
All-Sky Solar Radiation	RadAvg	1439	1676	1858	1954	1881	1914	1940	1891	1785	1594	1399	1345	
	RadStd	49	37	76	109	129	111	98	64	115	105	102	61	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.59	+1.06	N/A	N/A	N/A	N/A	N/A	+203	+202
(47) Regional (8 neighbors)		+0.50	+0.85	N/A	+0.59	N/A	N/A	N/A	+203	+202

Nomenclature: See separate page